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Here are the answers and detailed explanations for the four O Level Chemistry multiple-choice questions on **Topic 12: The Reactivity Series**.

Question 1

Text:

W, X, Y and Z are four different metals.

- W does not react with acids.
- X reacts with steam and acids but not with cold water.
- Y reacts with cold water.
- Z reacts slowly with acids but not with steam.

Which row shows the ease with which the metals can be extracted from their ores?

easiest to extract	most difficult to extract
A	W
B	W
C	Y
D	Y

Step-by-step reasoning

1. Reactivity order from given observations

- **Y** reacts with cold water → most reactive (group 1/2 metals like Ca).
- **X** reacts with steam (but not cold water) and acids → moderately reactive (e.g., Zn, Fe).

- **Z** reacts slowly with acids but not with steam → less reactive than X, but more reactive than W.
- **W** does not react with acids → least reactive (e.g., Cu, Ag, Au).

Reactivity order: **Y > X > Z > W**

2. Extraction ease vs. reactivity

- More reactive metals are **more difficult** to extract (require electrolysis).
- Less reactive metals are **easier** to extract (can be reduced by carbon or heat alone).

So:

- **Easiest to extract** = least reactive = **W**
- **Most difficult to extract** = most reactive = **Y**

3. Match with table

Easiest (W) → ... → Most difficult (Y)

Looking at options:

A: W X Z Y → W (easiest), Y (most difficult)  Correct order.

B: W Z X Y → W (easiest), Y (most difficult)  But Z more reactive than X? No, X > Z from data. Wrong.

C: Y X Z W → Y easiest  Wrong — Y is most reactive, so hardest to extract.

D: Y Z X W → Y easiest  Wrong.

Answer: A

Question 2

Text:

Some metals can be extracted from their ores by carbon.

Iron and metals less reactive than iron can be extracted from their ores by hydrogen.

Which row is correct?

extraction by carbon	extraction by hydrogen
A	calcium
B	calcium
C	copper
D	copper

Step-by-step reasoning

1. Extraction by carbon

- Carbon is used for metals **below aluminium** in reactivity series (Zn, Fe, Cu, etc.), but not for very reactive metals like Ca, Na, Mg (they need electrolysis).
- From given: "Some metals can be extracted by carbon" — common examples: **copper, iron, zinc, lead**.
- Calcium is too reactive — extracted by electrolysis, not carbon. So A and B wrong.

2. Extraction by hydrogen

- The statement: "Iron and metals less reactive than iron can be extracted by hydrogen."
- Less reactive than iron: Cu, Pb, Ag, etc.
- Zinc is **more reactive than iron** — so cannot be extracted by hydrogen in practice (hydrogen isn't a strong enough reducing agent for Zn).
- Lead is less reactive than iron — yes, can be extracted by H_2 (but H_2 is rarely used industrially for Pb; this is a theory Q).

3. Check options

C: Copper (carbon) ✓, Lead (hydrogen) ✓ (Pb less reactive than Fe)

D: Copper (carbon) ✓, Zinc (hydrogen) ✗ (Zn more reactive than Fe — should be carbon, not H_2)

Answer: C

Question 3

Text:

Metal Q reacts slowly with cold water to produce hydrogen gas.

Metal R reacts slowly with dilute hydrochloric acid to produce hydrogen gas.

When metal S is added to an aqueous salt of metal R, a displacement reaction takes place. Metal S does not react with water.

Metal T does not react with dilute hydrochloric acid.

What is the order of reactivity for Q, R, S and T (least reactive → most reactive)?

least reactive	most reactive
A	Q
B	Q
C	T
D	T

Step-by-step reasoning

1. **Q reacts slowly with cold water** → fairly high reactivity (e.g., Ca, but slower than Na).
2. **R reacts slowly with dilute HCl** → less reactive than Q (since Q reacts with cold water, stronger reducing agent).
3. **S displaces R from its salt** → S is more reactive than R.
S does not react with water → S is below Q in reactivity (since Q does react with water).
So order so far: $R < S$, and $S < Q$.
4. **T does not react with dilute HCl** → very low reactivity, below R (since R reacts slowly with HCl).
So T is least reactive.
5. **Final order (least reactive first):**
T (no reaction with acid) → R (slow with acid) → S (displaces R, no reaction with water) → Q (slow with water)

That matches: $T \rightarrow R \rightarrow S \rightarrow Q$

6. Check options

A: Q S R T ❌

B: Q R S T ❌

C: T R S Q ✅

D: T S R Q ❌ (S more reactive than R? Yes, but R before S in this — wrong order)

Answer: C

Question 4

Text:

Which metal is most easily extracted from its ore?

A copper

B iron

C magnesium

D potassium

Step-by-step reasoning

1. Extraction ease increases as reactivity decreases

- Potassium (most reactive) → needs electrolysis → hardest
- Magnesium (very reactive) → electrolysis → very hard
- Iron (less reactive) → blast furnace with carbon → moderately easy

- Copper (low reactivity) → can be extracted by heating alone or carbon → easiest

2. **Order of extraction ease** (easiest first):

Copper > Iron > Magnesium > Potassium

Answer: A (Copper)

Final Answers Summary

1. **A**
2. **C**
3. **C**
4. **A**

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You Asked



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Here are the answers and detailed explanations for **Questions 5, 6, and 7** from your O Level Chemistry worksheet on **Topic 12: The Reactivity Series**.

Question 5

Text:

W, X, Y and Z are four metals.

- W reacts violently with cold water.
- X does not react with cold water but reacts with steam.
- Y reacts slowly with cold water.
- Z does not react with hydrochloric acid.

Which row shows the order of reactivity of W, X, Y and Z?

most reactive	least reactive
A	W
B	W
C	Z
D	Z

Step-by-step reasoning

1. Interpret reactivity from observations

- **Violent reaction with cold water** → very high reactivity (e.g., K, Na) → **W** is most reactive.
- **Slow reaction with cold water** → still reactive, but less than W → **Y** is less reactive than W but more reactive than X.
- **No reaction with cold water, but reacts with steam** → moderate reactivity (e.g., Zn, Fe) → **X** is less reactive than Y.
- **No reaction with HCl** → very low reactivity (e.g., Cu, Ag) → **Z** is least reactive.

2. Final order (most → least reactive)

W → Y → X → Z

3. Match with options

- A: W Y X Z  exactly matches
- B: W X Y Z  (X less reactive than Y, so wrong order)
- C: Z X Y W  (Z least reactive, not most)
- D: Z Y X W  (Z least reactive, not most)

Answer: A

Question 6

Text:

Three metals, X, Y and Z, were heated separately with the oxides of four metals P, Q, R and S, to find the order of reactivity.

The results are shown in the table:

(✓ = reaction observed, ✗ = no reaction)

metal	P	Q	R	S
X	X	X	X	X
Y	✓	✓	X	✓
Z	X	✓	X	X

What is the order of the reactivity of the metals from the **least reactive** to the **most reactive**?

- A. $X \rightarrow Z \rightarrow Y$
- B. $Y \rightarrow X \rightarrow Z$
- C. $Y \rightarrow Z \rightarrow X$
- D. $Z \rightarrow Y \rightarrow X$

Step-by-step reasoning

1. Understanding the experiment

A metal reacts with another metal's oxide if the metal is **more reactive** than the metal in the oxide.

- Example: If Y reacts with oxide of P $\rightarrow$ Y is more reactive than P.

2. Work out reactivity of P, Q, R, S (not needed fully, but helpful)

Let's compare each metal (X, Y, Z) by how many oxides they reduce.

- **X** reacts with **none** (all X) $\rightarrow$ X is **least reactive** among X, Y, Z.
- **Y** reacts with P, Q, S (3 out of 4) $\rightarrow$ Y is **very reactive**.
- **Z** reacts with Q only (1 out of 4) $\rightarrow$ Z is between X and Y.

3. Order (least $\rightarrow$ most reactive)

X (least) $\rightarrow$ Z $\rightarrow$ Y (most)

Check:

- Z reacts with Q but not P or S $\rightarrow$ means $Z > Q$ in reactivity, but $Z < P$ and $Z < S$.
- Y reacts with P, Q, S $\rightarrow Y > P, Y > Q, Y > S \rightarrow$ Y is the most reactive.
- X reacts with none $\rightarrow$ X is the least.

4. Match options

- A: $X \rightarrow Z \rightarrow Y$  correct
- B: $Y \rightarrow X \rightarrow Z$ 
- C: $Y \rightarrow Z \rightarrow X$ 
- D: $Z \rightarrow Y \rightarrow X$ 

Answer: A

Question 7

Text:

The reactions of four metals are shown:

metal	cold water	steam	dilute acid
W	gentle reaction	vigorous	vigorous
X	vigorous	violent	violent
Y	no reaction	no reaction	no reaction
Z	vigorous	violent	explosive

What is the order of reactivity of the metals?

(most reactive → least reactive)

- A. $W \rightarrow X \rightarrow Z \rightarrow Y$
- B. $X \rightarrow Z \rightarrow Y \rightarrow W$
- C. $Y \rightarrow W \rightarrow Z \rightarrow X$
- D. $Z \rightarrow X \rightarrow W \rightarrow Y$

Step-by-step reasoning

1. Compare cold water reactions

- W, X, Z all react with cold water → more reactive than Y.
- W: gentle, X: vigorous, Z: vigorous → X and Z more reactive than W.

2. Compare steam reactions

- W: vigorous, X: violent, Z: violent → similar info.

3. Compare acid reactions

- W: vigorous, X: violent, Z: explosive → Z is most reactive (explosive with acid).

4. Compare X vs Z

Cold water: both vigorous — same.

Acid: X violent, Z explosive → $Z > X$.

So Z more reactive than X.

5. Compare W vs X

Cold water: W gentle, X vigorous → $X > W$.

6. Order (most → least)

Z (most) → X → W → Y (least)

7. Match options

A: W X Z Y ❌

B: X Z Y W ❌ (Y not least — Y is least)

C: Y W Z X ❌

D: Z X W Y ✅ exactly matches

Answer: D

Final Answers Summary

5. **A**

6. **A**

7. **D**

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You Asked



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Here are the answers and detailed explanations for **Questions 8, 9, 10, and 11** from your O Level Chemistry worksheet on **Topic 12: The Reactivity Series**.

Question 8

Text:

The reactivity series for some metals is shown:

most reactive	least reactive
Mg	Al

Which reaction will occur?

- A. aluminium oxide + iron $\rightarrow$ aluminium + iron(III) oxide
- B. copper(II) nitrate + silver $\rightarrow$ copper + silver nitrate
- C. iron(II) sulfate + copper $\rightarrow$ iron + copper(II) sulfate
- D. zinc sulfate + magnesium $\rightarrow$ zinc + magnesium sulfate

Step-by-step reasoning

A displacement reaction occurs **only if** the free metal is **more reactive** than the metal in the compound.

- **A:** $\text{Al oxide} + \text{Fe} \rightarrow \text{Al} + \text{Fe}_2\text{O}_3$
Fe less reactive than Al? No — Fe is below Al in series. So Fe cannot displace Al. ✗ No reaction.
- **B:** $\text{Cu}(\text{NO}_3)_2 + \text{Ag} \rightarrow \text{Cu} + \text{AgNO}_3$
Ag less reactive than Cu? Yes — Ag is below Cu. But Ag cannot displace Cu because Ag is **less reactive**. To displace Cu, you need a metal **above** Cu. ✗ No reaction.
- **C:** $\text{FeSO}_4 + \text{Cu} \rightarrow \text{Fe} + \text{CuSO}_4$
Cu less reactive than Fe? Yes — Cu is below Fe. Cu cannot displace Fe. ✗ No reaction.
- **D:** $\text{ZnSO}_4 + \text{Mg} \rightarrow \text{Zn} + \text{MgSO}_4$
Mg more reactive than Zn? Yes — Mg is above Zn in the series.
So Mg displaces Zn from ZnSO_4 . ✔ Reaction occurs.

Answer: D

Question 9

Text:

- Metal W reacts slowly with cold water to produce hydrogen gas.
- Metal X reacts slowly with dilute hydrochloric acid to produce hydrogen gas.
- When metal Y is added to the salt of metal X, a displacement reaction takes place.
- Metal Y does not react with water.
- Metal Z does not react with dilute hydrochloric acid.

What is the order of reactivity for W, X, Y and Z? (least reactive $\rightarrow$ most reactive)

least reactive	most reactive
A	W
B	Y
C	Z

least reactive	most reactive
D	Z

Step-by-step reasoning

1. **W** reacts slowly with cold water → high reactivity (e.g., Ca).
2. **X** reacts slowly with HCl (but no mention of cold water) → less reactive than W.
3. **Y** displaces X from X's salt → Y is **more reactive** than X.
Y does **not** react with water → Y is **less reactive** than W (since W does react with water).
So: $X < Y < W$.
4. **Z** does not react with HCl → very low reactivity, below X.
So: $Z < X$.
5. **Final order (least → most reactive):**
 $Z \rightarrow X \rightarrow Y \rightarrow W$
6. **Match options**
 - A: W Y X Z ❌
 - B: Y X W Z ❌
 - C: Z X Y W ✅ exactly matches
 - D: Z Y X W ❌ (Y more reactive than X? Yes, but order is wrong — here X before Y is correct)

Answer: C

Question 10

Text:

Three metals, X, Y and Z, were heated separately with the oxides of four metals P, Q, R and S.
Results (✓ = reaction, ✗ = no reaction):

metal	P	Q	R	S
X	✗	✗	✗	✗
Y	✓	✓	✗	✓
Z	✗	✓	✗	✗

What is the order of reactivity from **most reactive** to **least reactive**?

- A. $X \rightarrow Z \rightarrow Y$
- B. $Y \rightarrow X \rightarrow Z$

C. $Y \rightarrow Z \rightarrow X$

D. $Z \rightarrow Y \rightarrow X$

Step-by-step reasoning

1. **X** reacts with no oxides $\rightarrow$ X is **least reactive** (cannot displace any metal).

2. **Y** reacts with P, Q, S (3 of 4) $\rightarrow$ Y is **most reactive** among X, Y, Z.

3. **Z** reacts with Q only $\rightarrow$ Z is between X and Y.

Order (most $\rightarrow$ least): **$Y \rightarrow Z \rightarrow X$**

4. Check options

A: $X \rightarrow Z \rightarrow Y$  (this is least $\rightarrow$ most)

B: $Y \rightarrow X \rightarrow Z$ 

C: $Y \rightarrow Z \rightarrow X$  correct

D: $Z \rightarrow Y \rightarrow X$ 

Answer: C

Question 11

Text:

Iron is often used as an alloy called stainless steel. Stainless steel does not easily rust.

(a) State the meaning of the term alloy.

Answer

An **alloy** is a **mixture** of a metal with one or more other elements (which may be metals or non-metals), that has been designed to have **improved properties** (such as greater strength, corrosion resistance, or hardness) compared to the pure metal.

For example, stainless steel is an alloy of iron, chromium, and nickel, which resists rusting.

Final Answers Summary

8. **D**

9. **C**

10. **C**

11. (a) An alloy is a mixture of a metal with one or more other elements to improve its properties (e.g., strength or corrosion resistance).

